

Standard Model Rules

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1 Standard Model

1.1 Electroweak-Quark Sector

We write the covariant terms in the lepton sector:

$$\mathcal{L}_{SM} \supset i\bar{q}^i \not{D} P_L q^j + i\bar{u}^i \not{D} P_R u^j + i\bar{d}^i \not{D} P_R d^j$$

The corresponding covariant derivatives will be considered,

$$(D_\mu - \partial_\mu) P_L q^i = -i \left(\frac{1}{6} g' B_\mu + g \frac{\sigma^a}{2} W_\mu^a + g_s \frac{\lambda^a}{2} G_\mu^a \right) P_L q^i,$$

$$(D_\mu - \partial_\mu) P_R u^i = -i \left(\left(\frac{2}{3} \right) g' B_\mu + g_s \frac{\lambda^a}{2} G_\mu^a \right) P_R u^i, \quad (D_\mu - \partial_\mu) P_R d^i = -i \left(\left(-\frac{1}{3} \right) g' B_\mu + g_s \frac{\lambda^a}{2} G_\mu^a \right) P_R d^i,$$

where the charges and the conventions they follow has been made explicit. Further, we make the following reparametrizations:

$$\begin{pmatrix} W_\mu^+ \\ W_\mu^- \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \begin{pmatrix} W_\mu^1 \\ W_\mu^2 \end{pmatrix}, \quad \begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \cos \theta_W & \sin \theta_W \\ -\sin \theta_W & \cos \theta_W \end{pmatrix} \begin{pmatrix} B_\mu \\ W_\mu^3 \end{pmatrix},$$

where θ_W is the Weinberg angle that satisfies $\tan \theta_W = g'/g$. It is now possible to read off the gauge interactions of the SM leptons. However, to be more explicit, note that:

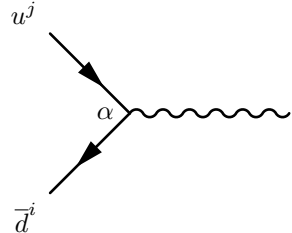
$$\frac{\delta^2 \mathcal{L}_{SM}}{\delta \bar{q}_L^i \delta q_L^i} \supset \frac{g}{2} (\sigma^1 W^1 + \sigma^2 W^2) = \frac{g}{2} \begin{pmatrix} 0 & (W^1 - iW^2) \\ (W^1 + iW^2) & 0 \end{pmatrix} = \frac{g}{\sqrt{2}} \begin{pmatrix} 0 & W^+ \\ W^- & 0 \end{pmatrix}$$

To check that these do possess the correct charge, note that $YW^{1,2} = 0$ for the hypercharge operator Y , while $\tau_3 W^+ = \epsilon_{123} W^+$ and $\tau_3 W^- = \epsilon_{213} W^-$. Since $Q_{EM} = \tau_3 + Y$, these bosons are charged as expected.

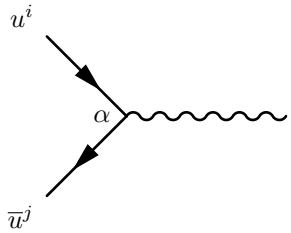
Similarly, one may consider their neutral counterparts:

$$\frac{\delta^2 \mathcal{L}_{SM}}{\delta \bar{q}_L^i \delta q_L^i} \supset (Q_Y g' \not{B} + Q_L g \sigma^3 \not{W}^3) = ((Q_Y g' c_W + Q_L g s_W \sigma^3) \not{A} + (-Q_Y g' s_W + Q_L g \sigma^3 c_W) \not{Z})$$

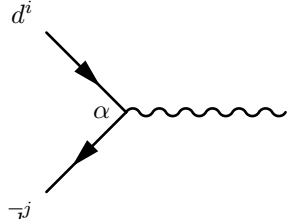
With the redefinition $e = g s_W = g' c_W$, one may now read off the necessary Feynman rules with the CKM matrix, $V_{CKM} = V_{L_u}^\dagger V_{L_d}$:



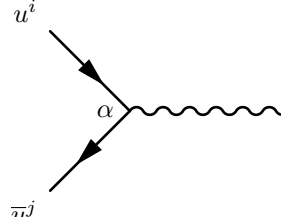
$$= i \frac{g}{\sqrt{2}} V_{CKM}^{ji*} \gamma^\alpha P_L = i \frac{e}{\sqrt{2} s_W} V_{CKM}^{ji*} \gamma^\alpha P_L, \quad (1)$$



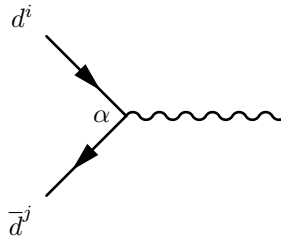
$$Z = -i \gamma^\alpha \delta_{ij} \left(\left(\frac{g' s_W}{6} - \frac{g c_W}{2} \right) P_L + \frac{2 g' s_W}{3} P_R \right) = -ie \gamma^\alpha \left(\frac{\tan \theta_W - 3 \cot \theta_W}{6} P_L + \frac{2 \tan \theta_W}{3} P_R \right), \quad (2)$$



$$Z = i\gamma^\alpha \delta_{ij} \left(\left(-\frac{g' s_W}{6} - \frac{g c_W}{2} \right) P_L + \frac{g' s_W}{3} P_R \right) = ie\gamma^\alpha \left(-\frac{\tan \theta_W + 3 \cot \theta_W}{6} P_L + \frac{\tan \theta_W}{3} P_R \right), \quad (3)$$



$$\gamma = i\gamma^\alpha \delta_{ij} \left(\left(\frac{g' c_W}{6} + \frac{g s_W}{2} \right) P_L + \frac{2g' c_W}{3} P_R \right) = \frac{2}{3} ie\gamma^\alpha, \quad (4)$$



$$\gamma = i\gamma^\alpha \delta_{ij} \left(\left(\frac{g' c_W}{6} - \frac{g s_W}{2} \right) P_L - \frac{g' c_W}{3} P_R \right) = -\frac{1}{3} ie\gamma^\alpha. \quad (5)$$

If one wishes to compare these vertices to [1], it is important to note that the covariant derivative follows a different sign convention regarding charges. A combination of trigonometric and algebraic identities aids us further with this comparison:

$$\frac{-\tan \theta_W \pm 3 \cot \theta_W}{6} P_L - \frac{1 \pm 3}{6} \tan \theta_W P_R = \pm \frac{1}{2s_W c_W} \left(P_L - (1 \pm \frac{1}{3}) s_W^2 \right)$$

1.2 Lepton-Gauge Sector

We write the covariant terms in the lepton sector:

$$\mathcal{L}_{SM} \supset i\bar{\ell}^i \not{D} P_L \ell^i + i\bar{e}^i \not{D} P_R e^i$$

The corresponding covariant derivatives will be considered,

$$(D_\mu - \partial_\mu) P_L \ell^i = -i \left(\left(-\frac{1}{2} \right) g' B_\mu + g \frac{\sigma^a}{2} W_\mu^a \right) P_L \ell^i, \quad (D_\mu - \partial_\mu) P_R e^i = -i ((-1) g' B_\mu) P_R e^i,$$

where the charges and the conventions they follow has been made explicit. Further, we make the following reparametrizations:

$$\begin{pmatrix} W_\mu^+ \\ W_\mu^- \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \begin{pmatrix} W_\mu^1 \\ W_\mu^2 \end{pmatrix}, \quad \begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \cos \theta_W & \sin \theta_W \\ -\sin \theta_W & \cos \theta_W \end{pmatrix} \begin{pmatrix} B_\mu \\ W_\mu^3 \end{pmatrix},$$

where θ_W is the Weinberg angle that satisfies $\tan \theta_W = g'/g$. It is now possible to read off the gauge interactions of the SM leptons. However, to be more explicit, note that:

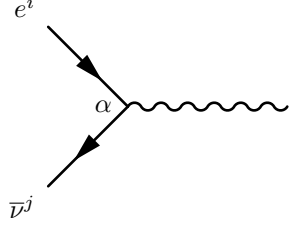
$$\frac{\delta^2 \mathcal{L}_{SM}}{\delta \bar{\ell}_L^i \delta \ell_L^i} \supset \frac{g}{2} (\sigma^1 W^1 + \sigma^2 W^2) = \frac{g}{2} \begin{pmatrix} 0 & (W^1 - iW^2) \\ (W^1 + iW^2) & 0 \end{pmatrix} = \frac{g}{\sqrt{2}} \begin{pmatrix} 0 & W^+ \\ W^- & 0 \end{pmatrix}$$

To check that these do possess the correct charge, note that $YW^{1,2} = 0$ for the hypercharge operator Y , while $\tau_3 W^+ = \epsilon_{123} W^+$ and $\tau_3 W^- = \epsilon_{213} W^-$. Since $Q_{EM} = \tau_3 + Y$, these bosons are charged as expected.

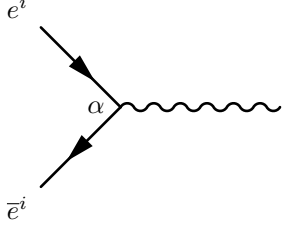
Similarly, one may consider their neutral counterparts:

$$\frac{\delta^2 \mathcal{L}_{SM}}{\delta \bar{\ell}_L^i \delta \ell_L^i} \supset \frac{1}{2} (-g' \not{B} + g \sigma^3 W^3) = \frac{1}{2} ((-g' \cos \theta_W + g \sin \theta_W \sigma^3) \not{A} + (g' \sin \theta_W + g \sigma^3 \cos \theta_W) \not{Z})$$

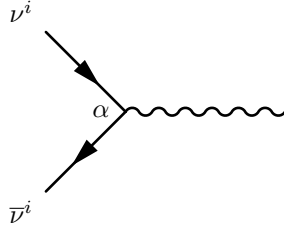
With the redefinition $e = g \sin \theta_W = g' \cos \theta_W$, one may now read off the necessary Feynman rules that are clearly flavor universal and flavor diagonal:



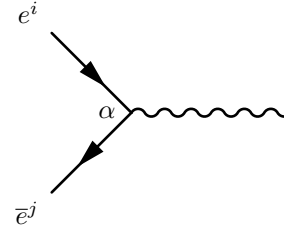
$$W = i \frac{g}{\sqrt{2}} \delta_{ij} \gamma^\alpha P_L = i \frac{e}{\sqrt{2} \sin \theta_W} \delta_{ij} \gamma^\alpha P_L, \quad (6)$$



$$Z = i \gamma^\alpha \left(\frac{g' \sin \theta_W - g \cos \theta_W}{2} P_L + g' \sin \theta_W P_R \right) = i e \gamma^\alpha \left(\frac{\tan \theta_W - \cot \theta_W}{2} P_L + \tan \theta_W P_R \right), \quad (7)$$



$$Z = i \gamma^\alpha \frac{g' \sin \theta_W + g \cos \theta_W}{2} P_L = i e \gamma^\alpha \frac{\tan \theta_W + \cot \theta_W}{2} P_L, \quad (8)$$



$$\gamma = i \gamma^\alpha \delta_{ij} \left(-\left(\frac{g' c_W}{2} + \frac{g s_W}{2} \right) P_L - \frac{g' c_W}{2} P_R \right) = -i \delta_{ij} e \gamma^\alpha, \quad (9)$$

If one wishes to compare these vertices to [1], it is important to note that the covariant derivative follows a different sign convention regarding charges. A combination of trigonometric and algebraic identities aids us further with this comparison:

$$\frac{\tan \theta_W - \cot \theta_W}{2} P_L + \tan \theta_W P_R = -\frac{1}{2 \sin \theta_W \cos \theta_W} P_L + \tan \theta_W$$

1.3 Goldstones

The Goldstones arise from a reparametrization of the BEH doublet that produces the Standard Model Higgs, upon acquiring a non-zero vacuum expectation value, v . Expressed as a relic of its status as a doublet under $SU(2)_L$ along with the choice of an arbitrary unit vector,

$$\phi(x) = \frac{v}{\sqrt{2}} e^{-i 2 \pi^i(x) \frac{\sigma^i}{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \left(1 + \frac{\rho(x)}{v} \right). \quad (10)$$

The interactions of the Goldstone bosons are entirely determined by the interactions and symmetries of the BEH field ϕ .

1.3.1 Yukawa Interactions

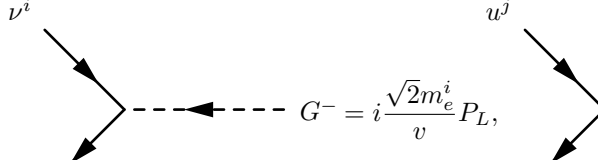
The Yukawa interactions of the Standard Model may be written as:

$$\mathcal{L}_{Yuk} = Y_u^{ij} \bar{q}_L^i \tilde{\phi} u_R^j + Y_d^{ij} \bar{q}_L^i \phi d_R^j + Y_e^{ij} \bar{\ell}_L^i \phi e_R^j + \text{h.c.} \quad (11)$$

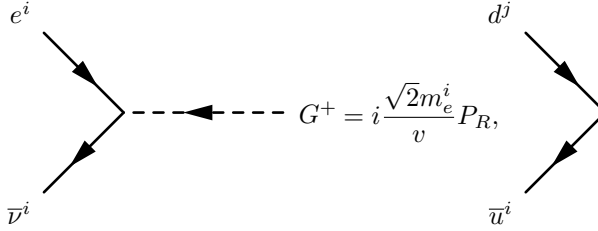
Past the Electroweak symmetry breaking scale, these fields may be re-expressed in terms of their constituents in a sensible fashion. It is useful to work with the infinitesimal expansion of the exponential.

$$\begin{aligned}\mathcal{L}_{Yuk}^{quark} = & Y_u^{ij} \left(\frac{i(\pi^1 + i\pi^2)(\rho + v)}{\sqrt{2}v} \bar{d}_L^i + \frac{i(\pi^3 + iv)(\rho + v)}{\sqrt{2}v} \bar{u}_L^i \right) u_R^j + Y_d^{ij} \left(\frac{\left(1 + \frac{i\pi^3}{v}\right)v \left(\frac{\rho}{v} + 1\right)}{\sqrt{2}} \bar{d}_L^i - \frac{i(\pi^1 - i\pi^2) \left(\frac{\rho}{v} + 1\right)}{\sqrt{2}} \bar{u}_L^i \right) d_R^j \\ & + Y_e^{ij} \left(\frac{v \left(1 + \frac{i\pi^3}{v}\right) \left(\frac{\rho}{v} + 1\right)}{\sqrt{2}} \bar{e}_L^i - \frac{i(\pi^1 - i\pi^2) \left(\frac{\rho}{v} + 1\right)}{\sqrt{2}} \bar{\nu}_L^i \right) e_R^j + \text{h.c.}\end{aligned}$$

Herein, it is convenient to consider the charged Goldstones $G^\pm = \frac{\pi^1 \pm i\pi^2}{\sqrt{2}}$, and the neutral $G^0 = \pi^3$. Note that in going to the mass eigenbasis of the fermions (ignoring the PMNS and any RH neutrinos), one gains factors of CKM matrix elements, $V_{\text{CKM}} = V_{L_u}^\dagger V_{L_d}$. Thus, we may now read off the Feynman rules for the charged Goldstones:



$$G^- = i \frac{\sqrt{2}m_e^i}{v} P_L, \quad G^- = i V_{\text{CKM}}^{ji*} \left(\frac{\sqrt{2}m_d^i}{v} P_L + \frac{\sqrt{2}m_u^j}{v} P_R \right), \quad (12)$$



$$G^+ = i \frac{\sqrt{2}m_e^i}{v} P_R, \quad G^+ = i V_{\text{CKM}}^{ij} \left(\frac{\sqrt{2}m_u^i}{v} P_L + \frac{\sqrt{2}m_d^j}{v} P_R \right), \quad (13)$$

References

- [1] Janusz Rosiek. Complete set of Feynman rules for the MSSM: Erratum. 11 1995.